

1 Performance Benchmarks

Performance Comparison Report

Advanced Framework Benchmarks across CPU and GPU Architectures

1.1 LAMMPS vs. MCCB-gpu (Large-Scale Scales)

Benchmarks were conducted using two distinct NVIDIA hardware architectures: the **RTX 2000** and the **RTX 4500**.

Table 1: Simulation Runtime (Seconds) by Particle Count (N) for Large Scales

Particle Count (N)	LAMMPS Time		MCCB-gpu Time	
	RTX 2000	RTX 4500	RTX 2000	RTX 4500
492 450	7.0	5.0	18.0	3.8
1 969 920	22.0	18.0	66.0	13.2
3 939 840	44.0	34.0	123.0	25.6

1.2 gpMC (CPU) vs. MCCB-gpu (GPU) Scaling (NVT and NpT Ensembles)

This benchmark assesses execution runtime across both canonical (NVT) and isobaric-isothermal (NpT) ensembles, scaling from 8000 to 125 000 particles for a binary Lennard-Jones mixture ($N_1 : N_2 = 2 : 1$, $T^* = 3.0$, $r_c = 2.5\sigma$, with isotropic volume trial moves executed every cycle in NpT). We compare the CPU-driven implementation (**gpMC**, compiled with Intel Fortran **ifx** 2025.2.0) executed on an **Intel(R) Xeon(R) Gold 6548Y+** (32 cores / 64 threads, 2.20 GHz) against GPU acceleration (**MCCB-gpu** Version 2.8.0) on an **NVIDIA RTX PRO 4500 Blackwell** (32 GB VRAM) for workloads of 100 Monte Carlo sweeps.

Trial Displacements per Particle Normalization. A critical distinction between the two codes lies in the operational definition of a Monte Carlo cycle:

- In **gpMC**, each cycle attempts exactly one single-particle displacement per particle ($\mu_{\text{CPU}} = 1.0$). Over 100 cycles, exactly 100 trial moves per particle are evaluated ($100 \times N$ total moves).
- In **MCCB-gpu**, the GPU checkerboard decomposition attempts $N_{\text{move}} = 125$ trial displacements per active cell across the 8 sublattices during each cycle, yielding a total of $N_{\text{trans}} = N_{\text{cells}} \times N_{\text{move}}$ moves per cycle. The number of trial displacements attempted *per particle per cycle* is therefore:

$$\mu_{\text{GPU}} = \frac{N_{\text{cells}} \times N_{\text{move}}}{N} \quad (1)$$

yielding $\mu_{\text{GPU}} = 3.375, 1.728, 1.953$, and 2.744 moves/particle/cycle for $N = 8000, 15\,625, 32\,768$, and $125\,000$, respectively.

Consequently, over 100 simulation sweeps, **MCCB-gpu** evaluates between $1.73\times$ and $3.38\times$ **more trial displacements per particle** than **gpMC**. To provide an unbiased comparison of actual Monte Carlo sampling throughput, Table 2 reports both the raw measured wallclock time t_{raw} and the normalized GPU time $t_{\text{norm}} = t_{\text{raw}}/\mu_{\text{GPU}}$ (scaled to an equivalent workload of 100 moves per particle), alongside the corresponding effective speedup:

$$S_{\text{effective}} = \frac{t_{\text{CPU}}}{t_{\text{norm, GPU}}} = S_{\text{raw}} \times \mu_{\text{GPU}} \quad (2)$$

Table 2: Simulation Runtime (Seconds) and Speedup across CPU and GPU Architectures (NVT and NpT Ensembles), Accounting for Differing Trial Displacements per Particle

Particles (N)	Moves/Part.	gpMC CPU Time (s)		MCCB-gpu GPU Time (s)		Effective Speedup	
	($\mu_{\text{GPU}} / \mu_{\text{CPU}}$)	NVT	NpT	NVT (Norm) [†]	NpT (Norm)	NVT	NpT
8000	3.375 / 1.0	9.95	15.16	4.97 (1.47)	6.26 (1.85)	6.76 $\times$ (2.00 $\times$)	8.17 $\times$ (2.42 $\times$)
15 625	1.728 / 1.0	23.59	35.95	8.73 (5.05)	11.88 (6.88)	4.67 $\times$ (2.70 $\times$)	5.23 $\times$ (3.03 $\times$)
32 768	1.953 / 1.0	38.63	59.23	8.05 (4.12)	9.94 (5.09)	9.37 $\times$ (4.80 $\times$)	11.64 $\times$ (5.96 $\times$)
125 000	2.744 / 1.0	175.91	254.09	7.86 (2.86)	10.14 (3.70)	61.41 $\times$ (22.39 $\times$)	68.76 $\times$ (25.06 $\times$)

Measured on node **ladon28** (Intel Xeon Gold 6548Y+ vs NVIDIA RTX PRO 4500 Blackwell) for 100 MC sweeps. In **gpMC**, each sweep attempts exactly 1.0 displacement per particle ($\mu_{\text{CPU}} = 1.0$). In **MCCB-gpu**, checkerboard decomposition evaluates $N_{\text{cells}} \times N_{\text{move}}$ trial displacements per sweep, yielding $\mu_{\text{GPU}} = N_{\text{cells}}N_{\text{move}}/N$ trial moves per particle (3.375, 1.728, 1.953, and 2.744 moves/particle/cycle for $N = 8000, 15\,625, 32\,768, 125\,000$, respectively). The normalized GPU runtime $t_{\text{norm}} = t_{\text{raw}}/\mu_{\text{GPU}}$ (in parentheses) reflects the execution time scaled to an equivalent 1.0 move/particle workload (100 moves per particle total). Effective speedup is defined as $S_{\text{eff}} = t_{\text{CPU}}/t_{\text{norm}} = S_{\text{raw}} \times \mu_{\text{GPU}}$, with raw unadjusted cycle speedup in parentheses. [†] Pure GPU kernel loop times (excluding I/O) on RTX 4500 are 0.85 s ($N = 8000$), 3.20 s ($N = 15\,625$), 2.61 s ($N = 32\,768$), and 1.93 s ($N = 125\,000$); on RTX 2000: 1.21 s, 4.27 s, 3.49 s, and 4.34 s.

1.3 Thermodynamic Consistency Benchmarks ($NpT \leftrightarrow NVT$)

To verify the physical fidelity and thermodynamic equivalence of the simulation ensembles, an automated two-stage workflow was developed and executed across cluster nodes **ladon28** and **ladon29**:

1. **Stage 1** (NpT): An isobaric-isothermal simulation is initialized and equilibrated at target pressure P_{target}^* and temperature T^* , recording the mean volume $\langle V \rangle_{NpT}$ (equilibrium density $\langle \rho^* \rangle_{NpT}$) and generating an exact configuration restart checkpoint.
2. **Stage 2** (NVT): A canonical simulation is executed from the restart configuration at fixed density $\rho^* = \langle \rho^* \rangle_{NpT}$.
3. **Validation**: Statistical equivalence $\langle U/N \rangle_{NVT} \approx \langle U/N \rangle_{NpT}$ is verified for continuous potential fluids (Lennard-Jones and Site-Site Patchy colloids), while the virial contact extrapolation $\langle P_{\text{virial}} \rangle$ is cross-validated against the analytical Carnahan-Starling equation of state for Hard Spheres.

Simulations comprised $N = 7695$ particles per system, executed concurrently under SLURM on dedicated RTX PRO 4500 GPUs.

Table 3: Thermodynamic Consistency Validation across Interaction Models

Model	Observable	NpT Ensemble	NVT Ensemble	Discrepancy	Agreement
Lennard-Jones ($T^* = 1.50, P^* = 0.60$)	Density $\langle \rho^* \rangle$	0.47230 ± 0.00405	0.47029 (fixed)	0.43%	0.50σ
	Energy $\langle U/N \rangle$	-2.64028 ± 0.02328	-2.63133 ± 0.01100	0.34%	0.35σ (Z-score)
Hard Spheres ($T^* = 1.00$)	Virial P_{NpT}^*	0.67674 ± 0.00218	–	0.03%	vs. $P_{CS} = 0.67697$
	Virial P_{NVT}^*	–	0.72284 ± 0.00059	0.04%	vs. $P_{CS} = 0.72311$
Site-Site Patchy ($T^* = 0.154, P^* = 0.05$)	Density $\langle \rho^* \rangle$	0.71579 ± 0.00627	0.72375 (fixed)	1.11%	Equilibrated
	Energy $\langle U/N \rangle$	-1.87301 ± 0.00981	-1.88494 ± 0.00325	0.64%	1.15σ (Z-score)
	Restart Point U/N	-1.88562	-1.88560 ± 0.00034	0.001%	Exact Match

Evaluated for Hard Spheres (HS), Lennard-Jones (LJ), and Site-Site Patchy (SSP, 4-patch tetrahedral Palaiia model) fluids with $N = 7695$ particles. Virial pressure for HS is calculated via pair-distribution contact extrapolation $g(\sigma^+)$ and compared against the Carnahan-Starling equation of state $P_{CS}(\rho)$.

2 Discussion and Structural Insights

An analysis of the benchmarking profiles yields three key architectural and physical insights:

- **Heterogeneous Paradigm Shifts and Sampling Throughput:** Transitioning from multicore CPU architectures (gpMC) to modern GPU acceleration (MCCB-gpu on RTX PRO 4500) delivers substantial speedups. On a raw cycle-by-cycle basis, GPU speedups reach **$22.4\times$** (NVT) and **$25.1\times$** (NpT) at $N = 125\,000$. When normalizing for the actual number of trial displacements attempted per particle (since MCCB-gpu performs up to $3.38\times$ more trial moves per particle per cycle), the true sampling throughput speedup reaches **$61.4\times$** in the canonical (NVT) ensemble and **$68.8\times$** in the isobaric-isothermal (NpT) ensemble.
- **Anomalous Scaling Profile across Ensembles:** In both the NVT and NpT datasets, scaling from $N = 15\,625$ to $N = 125\,000$ exhibits an execution time reduction/plateau on the GPU (e.g., dropping from 8.73 s to 7.86 s in NVT , and plateauing at ~ 10 s in NpT , despite an $8\times$ particle increase). This demonstrates a hardware architectural threshold where GPU block-level memory coalescing and thread warp occupancy reach peak efficiency.
- **Thermodynamic Rigor and Conservation:** The two-stage consistency test suite confirms that MCCB-gpu preserves strict thermodynamic equivalence across ensembles: continuous models (LJ and SSP) reproduce canonical energies within sub-percent statistical uncertainty (0.34% for LJ, 0.64% for SSP, $Z < 1.2\sigma$), while athermal contact virial evaluation reproduces the analytical Carnahan-Starling equation of state within **$< 0.04\%$** accuracy.